

Response to comments from Reviewer 1

This paper sets out to develop a reliable method for monitoring internet and mobile phone adoption by gender and digital gender gaps at the subnational level in low- and middle- income countries. The need for this type of monitoring system is well motivated; digital technology adoption is associated with a wide array of improved development outcomes and digital platforms are increasingly used to target anti-poverty interventions. I commend the authors for taking on this effort. However, the data product they generate falls short of achieving their goal. For one, it provides only a single snapshot in time, which is likely already outdated. Furthermore, the estimates are not convincingly validated, errors are not well characterized, and the measures of uncertainty the authors produce leave much to be desired. These concerns all cast doubt on the usefulness of these data to practitioners. Methodologically, there is not anything novel here.

Major comments

1. The authors motivate the research project by arguing that reliable, subnational estimates of digital technology adoption by gender are essential for monitoring gender gaps within countries, tracking progress towards the global sustainable development goals, and implementing targeted development interventions. However, the estimates that they produce - a single snapshot of administrative level 1 digital mobile phone and internet adoption and digital gender gaps - does not seem well-suited to any of these tasks.

a. There is no time variation in the estimates and the authors provide no suggestion that their methodology could be used to track changes in digital technology adoption or digital gender gaps over time. Because many of the features are cross-sectional, I would be surprised if the method can produce skillful predictions of changes over time.

b. The authors train and evaluate their algorithm on ground-truth labels dating back to 2015, and two-thirds of the DHS surveys they rely on come from 2018 or earlier. This means that the estimates they produce are already quite outdated. Although 5-10 years may not make a huge difference for some development indicators, it does for fast-moving digital technologies. The World Bank estimates that the number of internet users in low- and middle-income countries increased by more than 80% over this period, from 31% of the population in 2015 to 57% in 2021. Mobile phone subscriptions in low- and middle-income increased by 15%, from 90 to 104 subscriptions per 100 individuals from 2015 to 2022.

c. While ADM1 estimates do represent a significant improvement over national estimates, are they really spatially granular enough to help target development interventions?

2. There is not nearly enough validation and error analysis in the paper.

a. Examining Figure 1 reveals that many of the prediction locations are far out of sample of the training data, both geographically and in digital technology adoption levels. For example, South America, North

Africa, and the Middle East hardly appear in the DHS training data but make up a substantial share of the prediction regions. Further, most of the training data has low digital technology adoption (below 50% for internet use), but many prediction regions seem to have very high adoption (above 75% for internet use). There are many reasons to suspect that a model trained almost exclusively on relatively poor countries in sub-Saharan Africa, South Asia, and Southeast Asia will transfer poorly to more affluent countries in South America, North Africa, and the Middle East. However, the authors currently only perform subnational validation exercises within the DHS sample. I therefore suspect that their validation measures overstate the full sample performance. Further, the text does not provide nearly enough caution about using these out-of-sample predictions.

b. There is virtually no error analysis included in the paper. The authors write in the Discussion "Future work could consider settings where predictions do better or worse and diagnose factors that explain these residuals." Understanding where predictions are better vs. worse is fundamental to their practical use and should not be left to future work.

c. The authors do provide a measure of uncertainty in Appendix C.2 (which visually confirms that out-of-sample predictions are of lower quality). Their measure of uncertainty is based on a model that regresses the residual against a set of all observable variables for each subnational unit with ground truth. But again, this is only within the training sample, so it likely underestimates out-of-sample uncertainty.

3. The predictions exhibit compressed variance relative to the ground truth measures. This is typical for machine learning predictions but may be particularly problematic in this setting if it causes the authors to systematically underestimate gender gaps. Relatedly, the paper's motivation focuses on gender gaps, but all the main text results show validation for digital adoption by gender only. Examining Appendix Figures A8 and A9 reveals that performance is systematically lower for gender gaps (R^2 of < 0.5 versus $R^2 > 0.7$).

Minor comments

1. The Pearson correlation coefficient is the only metric reported for the within-country performance. Is this the relevant measure for users? It captures relative ranking, but I imagine practitioners are more interested in absolute adoption levels.

2. How is adoption defined? This is never articulated in the manuscript.

3. The paper would benefit from more description of the features, especially the satellite-based geospatial data, which are never described (only listed without detail in Table A1).

Response to comments from Reviewer 2

Thank you for the opportunity to review this paper. The authors use an impressive set of methodologies to build subnational measures at the Area 1 level of access to internet and mobile phones by gender. They do this for an impressive set of 118 low and middle income countries (LMIC) covering 2,158 regions. They find large disparities internet and mobile usage among men and women, resulting in 385m fewer women

owning a mobile phone and 360m fewer using the internet globally. These are important findings and while the magnitude of the gap is important it is not altogether surprising given my priors.

The authors have produced an impressive set of data results, but I have concerns regarding the following

1) I found the organization to the paper to be lacking for a general interest journal. The paper ends abruptly at section 4, which is a section that is unnecessary in my opinion, without framing the context of their results. I wanted the authors to describe the gender gaps, especially to show that their data updates our priors about the gender gaps in mobile and internet access. Where are the gender gaps the largest by country? What are possible causes of these gender gaps? The authors provide the reader no context here.

2) Additionally, I was very interested in the within country disparities in access to internet. The authors show a few maps but do no across country comparison of say capital city to rural area breakdown of access to internet and mobile. Are these correlated with income? Or is it density of populations that are most predictive of these gaps? The reason this is important is that crafting policy to address these gaps will depend on the proximate cause, and the authors should point to possible ways to address these gaps, or at least suggest a reason they exist.

3) The methodology in general is sound, but I wonder whether the models are overfit or whether there are sufficient features to capture the global patterns desired. There's no training data in some regions, such as none in Latin America it seems, and limited in MENA. I wonder if this causes the high residuals in these countries. It's hard to tell the scale of A6 but if it's on raw percentage points this indicates models that are overfit to the training data.

4) I found myself wondering how surprising these results are. I'm not sure how they could test this, but perhaps compare these to existing surveys or methodology to show how much your approach updates our priors.

Minor concerns:

i) Figure A6 shows error rates correlated across countries, with some countries having much higher error rates than others. I think this is a consequence of the LOOCV approach (which I like, to be clear) but what it tells me is there is still a lot of country level variation in internet use that is uncaptured by the model. What could additionally capture this? Perhaps education?

ii) HDI seems to be a high predictor in many of these models, and generally HDI is constructed at the country level. Is that the case here or is it subnational?

iii) I really like the comparison of models in figures like A9. Really important to see the performance across models, and I very much like this level of transparency. The predicted/true scatterplots are great too. There's a lot of very nice work here.

Response to comments from Reviewer 3

This paper studies whether “digital gender gaps” in low-income countries – specifically, rates of adoption of mobile phone and internet technologies for men and women, and the ratio of adoption between the two genders – can be measured via social media data and other globally available information. The paper adopts a machine learning approach where the target of prediction is derived from individual-level data on mobile and internet adoption collected in DHS surveys and representative at the admin-1 (regional) level. The input features include data from Facebook’s Marketing API (primarily monthly active user counts disaggregated by gender at the admin-1 level), as well as “offline” features that are publicly available at the admin-1 level globally, including information on population (from WorldPop), development (from nighttime lights and gridded GDI/HDI indices), and internet speed (from the SpeedTest Global Index).

The main result presented in the paper is that mobile and internet adoption can be predicted quite accurately at the admin-1 level ($R^2 = 0.7-0.71$ for mobile adoption and $0.70-0.77$ for internet usage), even when using stringent leave-one-country-out cross validation for evaluation. The “digital gender gap” – the ratio of male adoption to female adoption of each technology – is more difficult to predict ($R^2 = 0.48$ for mobile ownership and 0.47 for internet access). Moreover, the paper demonstrates that the addition of Facebook advertising data substantially improves predictive accuracy over using “offline” features alone (R^2 improvement not directly reported in the text, but appears to be $0.1-0.2$ in a figure). Using the ML model’s predictions at the admin-1 level, the paper is able to demonstrate the utility of such predictions, for example drawing conclusions about the distribution of within- and between-country inequality in internet and mobile adoption.

I greatly enjoyed reading the paper and am grateful for the opportunity to review it. This work has a number of strengths: The paper is exceptionally well-written and clear, and I particularly appreciated the precision with which all of the data and machine learning methods were reported (and presumably implemented), with a large number of robustness checks in the supporting materials. I frequently found that questions that I had while reading the paper were later answered in further analysis. The work also systematically builds on past work on using Facebook advertising data (as well as other digital data sources) for mapping digital adoption and gender gaps at the national level (in particular Fatehkia et al. (2018)).

I have two major concerns about the paper that I think would be valuable to address before publication. The first relates to more clearly articulating the policy uses the authors see for these types of predictions, and ensuring that the data, evaluation metrics, and cross validation strategies support these downstream use cases. The second is a request for validation with external survey data. I go into detail on these two concerns below, and finish with a list of more minor concerns and suggestions for improvement.

Major concern #1: Policy uses and evaluation metrics. The introduction and discussion sections introduce (at least) four possible policy use cases the authors see for the predictions of gender-disaggregated mobile and internet adoption. I would like to see the paper more clearly identify these uses, and ensure that the data, evaluation metrics, and validation strategies used are consistent with the way the predictions would be used in practice. In some cases, this is simple (for example, articulating the policy use cases that 10-fold CV and leave-one-country-out CV are representative of); in other cases, they would require a great deal of additional data and analysis (for example, showing that the predictions can measure changes in digital adoption over time). While I don’t expect that the authors will include all these use cases in the

final version of the paper, I ask that they clearly articulate the use cases and only include those that are supported by the data and analysis. The four cases I identified, with commentary, are as follows:

1. Measurement of mobile/internet adoption and disparities in countries where survey data are available in only some areas: In this scenario, a country has survey data on digital adoption in some admin-1 areas, but not in all areas. In this use case, standard 10-fold cross validation is likely the most appropriate validation approach, because the model should be trained on some data from the country in question, in addition to data from other countries. Clarifying that this is the use case that 10-fold cross validation corresponds to, and motivating its relevance (are there any countries with this kind of spotty survey data?), would improve the clarity of the paper.

2. Measurement of mobile/internet adoption and disparities in countries where no survey data are available: This is a realistic scenario for countries that do not have any existing survey-based measures of digital adoption. The leave-one-country-out approach to validation is most consistent with this scenario, since the ML model should not be trained on any data from the country in question. Clarifying that this is the use case that LOCO-CV corresponds to, and motivating its relevance (how many countries lack admin-1 level data on digital adoption altogether?), would improve the clarity of the paper.

3. Fine-grained geographic targeting of development interventions: The introduction ends with: “As development programs are increasingly deployed via digital means, and are often targeted in local geographies, subnational estimates are critical to monitoring their progress and understanding how digital inequalities affect sustainable development” (page 4). If one of the main policy use cases for these predictions is the geographic targeting of interventions, it would be helpful to include evaluation metrics beyond R^2 that would help us understand how accurately these predictions can be used for targeting such interventions. What would be the inclusion and exclusion error rates if a policy aims to identify the areas with the least adoption or most inequality? Adding such metrics would improve the clarity and policy relevance of the paper, as they would allow readers to understand the utility of the adoption predictions in practice.

4. Measuring changes in digital adoption and digital gender gaps over time. Several parts of the paper hint that a particularly useful aspect of the ML-based predictions using Facebook user counts is that they may be able to measure changes in digital adoption over time, as Facebook user counts increase or decrease. For example, the significance statement suggests that the estimates could be used to “track progress toward sustainable development goals in real time” (page 1) and the discussion section includes that “estimates for all regions together will be publicly available and updated” (page 12-13). The ability to track changes in adoption over time would be immensely useful, since surveys like the DHS are typically conducted fairly infrequently. However, none of the results in the current version of the paper show that changes in digital adoption over time can be measured through the ML predictions – they only show that digital adoption at a single point in time can be measured with relatively high cross-sectional accuracy. I do think the paper would be greatly strengthened if it could show that the ML-based estimates can track changes in digital adoption over time, by gathering additional Facebook user counts and matching these input variables to multiple waves of longitudinal ground truth survey data. However, I imagine such a data collection effort might be out of scope for a revision of this paper. Instead, since the results do not support the ability to measure change over time, I would request that the authors remove these

suggestions from the introduction and discussion sections (of course, it would be great to highlight this as an important direction of future work).

Major concern #2: Lack of external validation. As the authors themselves acknowledge in the discussion section, “external comparisons with a high-quality, subnationally representative and gender-disaggregated census or survey would be a valuable validation of our estimates” (pages 13-14). However, I disagree with the authors that such survey data are not currently available. Unless I am mistaken, the Living Standards Measurement Surveys (LSMS) include such gender disaggregated data. Like the DHS, the LSMS surveys are enumerated across many countries; I did not check every country’s survey but did spot check the instruments for Malawi 2019 and Nigeria 2018: the Malawi survey collects individual-level data on phone ownership, and the Nigeria survey collects individual-level data on both phone ownership and internet access. The LSMS surveys, like the DHS, are representative at the admin-1 level. An external validation of the estimates with a handful of countries using the LSMS would greatly strengthen the paper’s results, showing that the ML model is not overfit to the DHS surveys, and providing a truly independent evaluation of the estimates.

In addition to these two major requests for revisions, I have a number of minor concerns and suggestions enumerated below:

1. One of the most interesting results of the paper is that while the input features are highly predictive of gender-disaggregated internet and mobile adoption ($R^2 = 0.70-0.77$), they are substantially less predictive of the digital gender gap ($R^2 = 0.47-0.48$). The authors write that this “reflects more uncertainty in our ground truth data” (page 7). It was not clear to me how this conclusion was drawn – how were other hypotheses ruled out? For example, I could imagine that adoption is more correlated with “offline” features like development, while the gender gap is less correlated with such features, and therefore more difficult to predict from the model inputs. I do not know that this is the case, but would like to understand how the authors ruled out this and other hypotheses besides noise in the ground truth data.

2. In many ways the most novel contribution of this paper and other related papers in the literature is the use of Facebook advertising data as an input feature to the model – it is not particularly surprising that the “offline” features, like population and development, are correlated with digital adoption, but it is exciting that is that the addition of social media data improves predictive accuracy. Therefore, from a narrative perspective, additional commentary on the difference between the online-only model and the offline-only model would be particularly helpful. In particular, Figure 5A very nicely shows the difference in R^2 between an offline-only and a Facebook-plus-offline model – the paper would be improved by articulating and digging into these differences in the text (that is, the fact that adding facebook features generally improves the R^2 by $\sim 0.1-0.2$ is as or more interesting to me than the raw $R^2 = 0.70-0.77$ of the combined model). Why might the gap between offline-only and offline-plus-Facebook accuracy be different for different measures? Also, why might offline features not improve over Facebook-only features for the Internet gender gap?

3. Many papers in the poverty prediction literature demonstrate the utility of digital data by providing predictions at a higher resolution than would be possible with surveys like the DHS, which are only representative at an admin-1 level (see e.g. Jean et al. (2016), Blumenstock et al. (2015), Chi et al. (2022)). Is it possible to obtain the Facebook input features at a higher geographic resolution, and thus to produce adoption predictions at a finer level than admin-1? If so, this would greatly strengthen the paper, as it would clearly demonstrate why these predictions are useful for measurement and geographic targeting, even in countries where traditional data are already available through surveys like the DHS.
4. Appendix C2 shows that the uncertainty in predictions is lower in low-adoption settings, which is also referenced in the main paper (page 7). If Figure A6 is measuring absolute error, would it not just be the case that prediction error in low-adoption errors is lower because the absolute values of these errors are smaller? An additional figure showing relative error would be informative.
5. The utility of the predictions would be better demonstrated if you included additional example uses of your predictions, expanding on the current section 2.2. For example, which country is most unequal? Which country is least unequal?
6. Supplementary materials appear to be missing a section on offline features (there is A1 on ground truth and A2 on facebook features, but no section on offline features). Table A1 is very helpful, though.
7. Inconsistent use of “machine-learning” vs. “machine learning” – my view is that the latter is standard.